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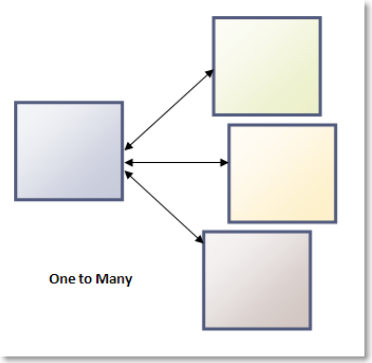
MULTI-TABLE QUERIES

Reading: Chapter 4
(pages 96-105)



ISE305 – DATABASE DESIGN & PRACTICE

SOME SLIDE MATERIAL FROM WATSON

Objectives

- Write queries for a database with a one-to-many relationship

Multi-table queries are similar to your single table queries – you just need to first join data from multiple tables to form a single table

Queries in a DB with Multiple Tables

- If related data exists in multiple tables
  1. Join the tables to form a new table
  2. Execute a SQL query on the joined table

Shares

ID	code	shareName	price	quantity	dividend	pe	nationID
1	FC	Freedonia Copper	$27.50	10529	$1.84	16	1
2	PT	Patagonian Tea	$55.25	12635	$2.50	10	1
3	AR	Abyssinian Ruby	$31.82	22010	$1.32	13	1
4	SLG	Sri Lankan Gold	$50.37	32868	$2.68	16	1
5	ILZ	Indian Lead & Zinc	$37.75	6390	$3.00	12	1
6	BE	Burmese Elephant	$0.07	154713	$0.01	3	1
7	BS	Bolivian Sheep	$12.75	231678	$1.78	11	1
8	NG	Nigerian Geese	$35.00	12323	$1.68	10	1
9	CS	Canadian Sugar	$52.78	4716	$2.50	15	1
10	ROF	Royal Ostrich Farms	$33.75	1234923	$3.00	6	1
12	MG	Minnesota Gold	$53.87	816122	$1.00	25	2
13	GP	Georgia Peach	$2.35	387333	$0.20	5	2
14	NE	Narembeen Emu	$12.34	45619	$1.00	8	3
15	QD	Queensland Diamond	$6.73	89251	$0.50	7	3
16	IR	Indooroopilly Ruby	$15.92	56147	$0.50	20	3
17	BD	Bombay Duck	$25.55	167382	$1.00	12	4

Nations

ID	code	nationName	exchangeRate
1	UK	United Kingdom	1.00
2	USA	United States	0.67
3	AUS	Australia	0.46
4	IND	India	0.02

Join

The dot notation refers to the nationID attribute in the Shares table

- Create a new table from two existing tables by matching on a common column

```
SELECT * FROM Shares, Nations
      WHERE Shares.nationID = Nations.ID;
```

Shares | Nations | SimpleJoin

Shares.ID	Shares.code	shareName	price	quantity	dividend	pe	nationID	Nations.ID	Nations.code	nationName	exchange
1	FC	Freedonia Copper	$27.50	10529	$1.84	16	1	1	UK	United Kingdom	1.00
2	PT	Patagonian Tea	$55.25	12635	$2.50	10	1	1	UK	United Kingdom	1.00
3	AR	Abyssinian Ruby	$31.82	22010	$1.32	13	1	1	UK	United Kingdom	1.00
4	SLG	Sri Lankan Gold	$50.37	32868	$2.68	16	1	1	UK	United Kingdom	1.00
5	ILZ	Indian Lead & Zinc	$37.75	6390	$3.00	12	1	1	UK	United Kingdom	1.00
6	BE	Burmese Elephant	$0.07	154713	$0.01	3	1	1	UK	United Kingdom	1.00
7	BS	Bolivian Sheep	$12.75	231678	$1.78	11	1	1	UK	United Kingdom	1.00
8	NG	Nigerian Geese	$35.00	12323	$1.68	10	1	1	UK	United Kingdom	1.00
9	CS	Canadian Sugar	$52.78	4716	$2.50	15	1	1	UK	United Kingdom	1.00
10	ROF	Royal Ostrich Farms	$33.75	1234923	$3.00	6	1	1	UK	United Kingdom	1.00
12	MG	Minnesota Gold	$53.87	816122	$1.00	25	2	2	USA	United States	0.67
13	GP	Georgia Peach	$2.35	387333	$0.20	5	2	2	USA	United States	0.67
14	NE	Narembeen Emu	$12.34	45619	$1.00	8	3	3	AUS	Australia	0.46
15	QD	Queensland Diamond	$6.73	89251	$0.50	7	3	3	AUS	Australia	0.46
16	IR	Indooroopilly Ruby	$15.92	56147	$0.50	20	3	3	AUS	Australia	0.46
17	BD	Bombay Duck	$25.55	167382	$1.00	12	4	4	IND	India	0.02

Relationships Summary

- 1:m relationship is provided with a primary key/foreign key pair
- Each foreign key must have a matching primary key
- Every primary key must be non-null
- Key pair must have the same type
- When modeling (ERD), we omit the foreign key attribute since it can be inferred from the relationship arrow

A common, but hard-to-find error when you are first starting

Join Example

You reference does not need to use the table name if the attribute name is unique

```
SELECT nationName, shareName, price, quantity, exchangeRate,
  price*quantity*exchangeRate AS stockValue
  FROM Shares,Nations
  WHERE Shares.nationID = Nations.ID
  ORDER BY nationName, shareName;
```

Query is performed on the joined table

To avoid name confusion, it is better to qualify the attribute name

Nations | Shares | JoinExample2

nationName	shareName	price	quantity	exchangeRate	stockValue
Australia	Indooroopilly Ruby	$15.92	56147	0.46	411175.7104
Australia	Narembeen Emu	$12.34	45619	0.46	258951.6916
Australia	Queensland Diamond	$6.73	89251	0.46	276303.2458
India	Bombay Duck	$25.55	167382	0.02	97506.71028
United Kingdom	Abyssinian Ruby	$31.82	22010	1.00	700358.2
United Kingdom	Bolivian Sheep	$12.75	231678	1.00	2953894.5
United Kingdom	Burmese Elephant	$0.07	154713	1.00	10829.91
United Kingdom	Canadian Sugar	$52.78	4716	1.00	248910.48
United Kingdom	Freedonia Copper	$27.50	10529	1.00	289547.5
United Kingdom	Indian Lead & Zinc	$37.75	6390	1.00	241222.5
United Kingdom	Nigerian Geese	$35.00	12323	1.00	431305
United Kingdom	Patagonian Tea	$55.25	12635	1.00	698083.75
United Kingdom	Royal Ostrich Farms	$33.75	1234923	1.00	41678651.25
United Kingdom	Sri Lankan Gold	$50.37	32868	1.00	1655561.16
United States	Georgia Peach	$2.35	387333	0.67	609855.8085
United States	Minnesota Gold	$53.87	816122	0.67	29456209.7338

Data Repetition Resulting From JOIN

- JOIN operation often results in data repetition
- GROUP BY clause is often used to summarize data

Shares | Nations | SimpleJoin

Shares.ID	Shares.code	shareName	price	quantity	dividend	pe	nationID	Nations.ID	Nations.code	nationName	exchange
1	FC	Freedonia Copper	$27.50	10529	$1.84	16	1	1	UK	United Kingdom	1.00
2	PT	Patagonian Tea	$55.25	12635	$2.50	10	1	1	UK	United Kingdom	1.00
3	AR	Abyssinian Ruby	$31.82	22010	$1.32	13	1	1	UK	United Kingdom	1.00
4	SLG	Sri Lankan Gold	$50.37	32868	$2.68	16	1	1	UK	United Kingdom	1.00
5	ILZ	Indian Lead & Zinc	$37.75	6390	$3.00	12	1	1	UK	United Kingdom	1.00
6	BE	Burmese Elephant	$0.07	154713	$0.01	3	1	1	UK	United Kingdom	1.00
7	BS	Bolivian Sheep	$12.75	231678	$1.78	11	1	1	UK	United Kingdom	1.00
8	NG	Nigerian Geese	$35.00	12323	$1.68	10	1	1	UK	United Kingdom	1.00
9	CS	Canadian Sugar	$52.78	4716	$2.50	15	1	1	UK	United Kingdom	1.00
10	ROF	Royal Ostrich Farms	$33.75	1234923	$3.00	6	1	1	UK	United Kingdom	1.00
12	MG	Minnesota Gold	$53.87	816122	$1.00	25	2	2	USA	United States	0.67
13	GP	Georgia Peach	$2.35	387333	$0.20	5	2	2	USA	United States	0.67
14	NE	Narembeen Emu	$12.34	45619	$1.00	8	3	3	AUS	Australia	0.46
15	QD	Queensland Diamond	$6.73	89251	$0.50	7	3	3	AUS	Australia	0.46
16	IR	Indooroopilly Ruby	$15.92	56147	$0.50	20	3	3	AUS	Australia	0.46
17	BD	Bombay Duck	$25.55	167382	$1.00	12	4	4	IND	India	0.02

GROUP BY Example

- Report by nation the total value of stock holdings

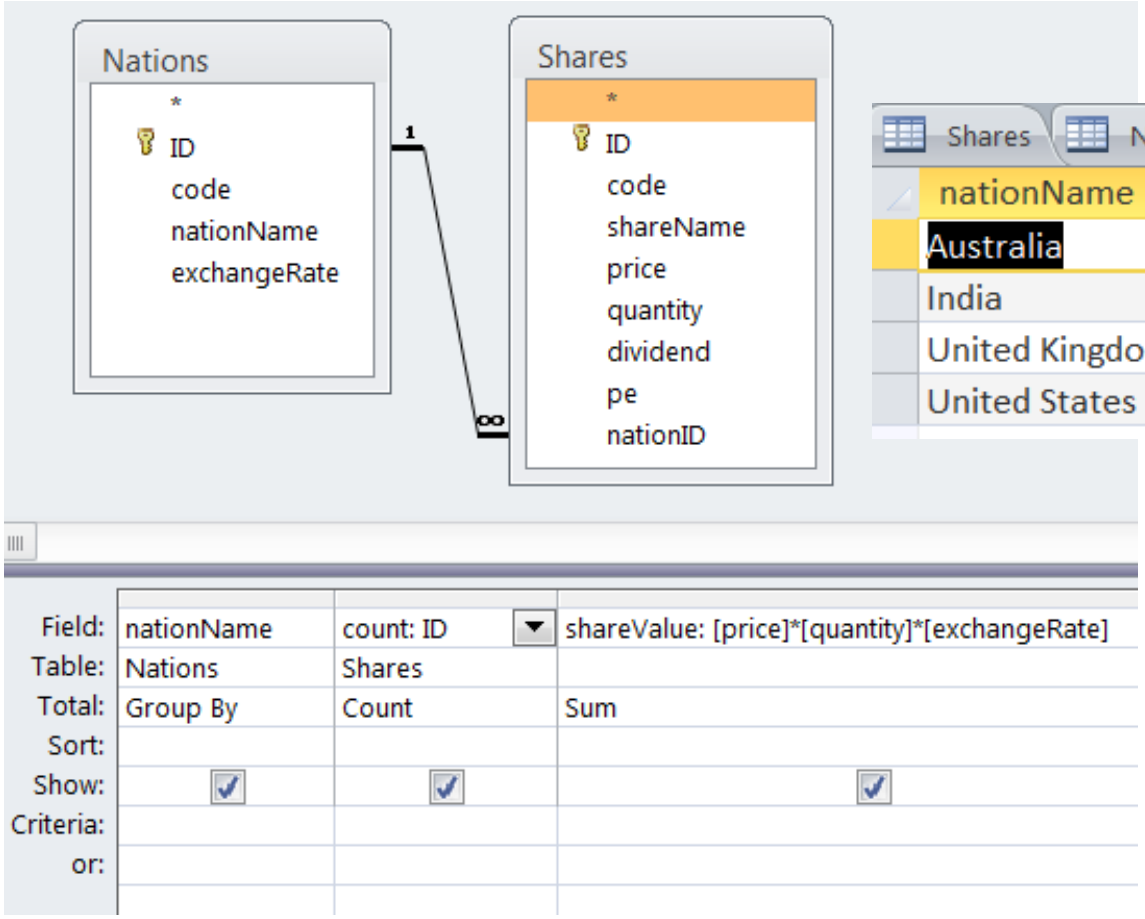
```
SELECT Nations.nationName, SUM(price*quantity*exchangeRate) AS
  shareValue
FROM Shares, Nations
WHERE Shares.nationID=Nations.ID
GROUP BY Nations.nationName;
```

Shares | Nations | GroupByExample

nationName	shareValue
Australia	946,430.65
India	97,506.71
United Kingdom	48,908,364.25
United States	30,066,065.54

JOIN Access Design View

- Number of stocks and their total value by nation



Shares | Nations | GroupByExample2

nationName	count	shareValue
Australia	3	946,430.65
India	1	97,506.71
United Kingdom	10	48,908,364.25
United States	2	30,066,065.54

Are We on Track?

- Report the total dividend payment by nation
- Report in British pounds
- Remember total dividends is shares multiplied by dividend per share

nationName	totalDividends
Australia	54,426.28
India	3,816.31
United Kingdom	4,338,465.91
United States	598,704.36

JOIN Notation

- SQL provides two syntactic approaches to specifying a join
  - Explicit join
  - Implicit join

```
FROM Nations INNER JOIN Shares
  ON Nations.ID = Shares.nationID
```

```
FROM Shares, Nations
WHERE Shares.nationID=Nations.ID
```

HAVING Example

- Report the total value of stocks (in pounds) for nations with two or more listed stocks

HavingExample

nationName	marketCapitalization
Australia	946,430.65
United Kingdom	48,908,364.25
United States	30,066,065.54

COUNT(*) will count the number of rows affected by the GROUP BY

```
SELECT Nations.nationName,
  SUM(price*quantity*exchangeRate)
   AS marketCapitalization
FROM Shares, Nations
WHERE Shares.nationID=Nations.ID
GROUP BY Nations.nationName
HAVING Count(*)>=2;
```

Subquery Example

- Report the names of all Australian stocks
- Note that the country name is in the Nations table and the stock name is in the Shares table

SubqueryExample

shareName
Narembeen Emu
Queensland Diamond
Indooroopilly Ruby

```
SELECT shareName
FROM  Shares
WHERE Shares.nationID IN
   (SELECT ID FROM Nations
     WHERE nationName = 'Australia') ;
```

Subquery returns a list (of size 1) of nation IDs

Alternate to Subquery

- The following SQL will obtain the same result, but the subquery approach might be more efficient

```
SELECT Shares.shareName
FROM Nations INNER JOIN Shares ON
  Nations.ID = Shares.nationID
WHERE Nations.nationName = 'Australia';
```

This approach might be more intuitive

Are We on Track?

- Report the average yield by nation (yield is defined as dividend divided by price, and is reported as a percentage)

nationName	Yield
Australia	6.22%
India	3.91%
United Kingdom	7.53%
United States	5.18%

Views - Virtual Tables

- An imaginary table constructed by the DBMS when required
- Only the definition of the view is stored, not the result

Not readily available in Access, so we will defer most of this until we get to MySQL

HW Assignment

1. For the previous Track, compute the weighted average (instead of the average) average yield by nation (yield is defined as dividend divided by price, and is reported as a percentage). The weighted average includes the quantity of shares as a weight
2. Report the minimum and maximum yield for each nation
3. Report the nations where the average yield of the shares exceeds the average yield of all shares

Did You Meet the Objectives?

- Write queries for a database with a one-to-many relationship